

# PISM — Physical Interaction Safety Module

**Parent Standard:** Physical Reality Interpretation Layer (PRIL™)  
**Category:** AI & Interpretation  
**Subcategory:** Physical Interaction Safety  
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**Compatibility:** OOF® Methodology OS™ · PRIL™ · CLIA® · RIS · INTEGROS® · EVIP® · SIMULOS®  
**Authority:** OOF®  
**Protection:** MIP® — Methodological Intellectual Property  
**Canonical Language:** English (UCL™)

## Canonical Definition

**Physical Interaction Safety Module** defines the methodological conditions under which autonomous systems interpret and restrict touch, contact, manipulation, collision risk, force application, and embodied interaction with humans, objects, or environments before physical execution proceeds.

A system satisfies PISM only if:

- physical interaction conditions are interpreted before contact or manipulation
- collision, touch, force, and contact risk are evaluated under defined conditions
- unsafe interaction is treated as an execution restriction condition
- uncertainty in contact conditions can stop, limit, or escalate execution
- physical action is limited when interaction safety interpretation is insufficient

A system that performs physical interaction without governed interaction safety interpretation does not satisfy PISM.

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## Module Function

PISM defines the physical-contact safety layer of embodied execution.

It ensures that systems do not treat touch, manipulation, collision avoidance, or contact behavior as purely mechanical outcomes.

The module applies wherever systems may touch, move, hold, push, carry, manipulate, approach into contact, or physically affect humans, objects, or surrounding environments.

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## Minimum Implementation Framework (MIF)

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## Step 1 — Define the Physical Interaction Boundary

The organization must define where physical interaction becomes operationally relevant.

Minimum requirement:

- contact boundary is explicit
  - interaction zones are distinguishable
  - undefined contact exposure is excluded from valid execution logic
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## Step 2 — Define Interaction Interpretation Conditions

The system must define how physical interaction conditions are interpreted.

Minimum requirement:

- contact conditions are not treated as raw mechanics only
  - touch, force, collision, and manipulation conditions are explicitly interpreted
  - uncertain interaction conditions are not silently treated as valid certainty
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## Step 3 — Define Safety Thresholds

The system must define what counts as safe, restricted, or unsafe interaction.

Minimum requirement:

- safety thresholds are explicit
  - force, contact, proximity-to-contact, and collision conditions are distinguishable
  - unsafe embodied interaction is identifiable before execution continues
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## Step 4 — Define Restriction and Escalation Logic

The system must define when interaction must slow, stop, restrict, release, or escalate.

Minimum requirement:

- restriction logic is explicit
  - unsafe or uncertain interaction can block physical action
  - continued contact under invalid safety conditions is not treated as valid
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## Step 5 — Preserve Safety Over Task Completion

The system must preserve physical safety above speed, grip success, throughput, or completion pressure.

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Minimum requirement:

- interaction risk overrides operational convenience
  - task urgency does not justify unsafe contact
  - physical safety remains structurally superior to execution success
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## Step 6 — Restrict Invalid Execution

The system must not be treated as valid if physical interaction proceeds while interaction safety interpretation is absent, degraded, or unreliable.

Minimum requirement:

- invalid interaction conditions are identifiable
  - degraded interaction interpretation blocks valid reliance
  - actuation alone does not restore safety validity
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## Use Case 1 — Robotic Manipulation Near Humans

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### Scenario

A robotic arm or embodied system manipulates tools, materials, or objects in an environment where humans may enter contact range during operation.

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### Application

PISM is used to ensure:

- the interaction boundary is defined
  - touch, collision, and force conditions are interpreted
  - unsafe contact risk triggers restriction
  - task execution does not override interaction safety
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### Result

The organization gains a stronger basis for preventing robotic manipulation from continuing when contact conditions are not sufficiently interpreted for safe operation.

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## Use Case 2 — Autonomous Service or Mobile Interaction System

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### Scenario

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A mobile system or service robot operates in a public or shared environment where it may contact people, objects, doors, carts, furniture, or other physical elements during movement or task execution.

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## Application

PISM is used to ensure:

- contact and manipulation conditions are interpreted before action
  - unsafe touch or collision risk is recognized
  - uncertain interaction conditions restrict execution
  - embodied behavior remains bounded by safety logic
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## Result

The organization gains a structurally governed physical interaction safety layer instead of relying on motion control alone as if it were enough for safe embodied operation.

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## Canonical Closing Statement

**If physical interaction is not interpreted for safety before contact, physical execution is not structurally valid.**

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## Related Documents

→ [Physical Reality Interpretation Layer \(PRIL™\)](#)

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## Canonical Source

<https://originopenfoundation.org/>